

API Automation Assessment – Data Consistency Across Multiple APIs

Objective

Validate that the posts returned by the API belong to the correct user and ensure data consistency between the **Users API** and **Posts API**.

User Story

As a QA Engineer, I want to verify that the posts returned for a user are associated with the correct **userId**, ensuring data integrity across multiple APIs.

API Flow

Step 1: Get All Users

Request

GET https://jsonplaceholder.typicode.com/users

Purpose

- Fetch the list of all users.
 - Dynamically select any one **userId** from the response.
-

Step 2: Get Posts for Selected User

Request

GET https://jsonplaceholder.typicode.com/posts?userId={userId}

Query Parameter

.queryParams("userId", userId)

Purpose

- Retrieve all posts belonging to the selected user.
-

Validations

1. Response Status

- Verify the API returns **HTTP Status Code 200**.
-

2. Response List Validation

- Verify the posts list is **not empty**.

Example:

```
assertFalse(posts.isEmpty());
```

3. User ID Validation

- Iterate through all returned posts.
- Verify every post contains the same `userId` used in the request.

Example:

```
for (Integer id : userIds) {  
    assertEquals(userId, id);  
}
```

Key Automation Concepts

JSON Parsing

- Parse the JSON response using **JsonPath**.
- Extract:
 - `userId`
 - List of posts

Example:

```
JsonPath jsonPath = response.jsonPath();
```

Query Parameter Handling

Pass the selected `userId` dynamically.

Example:

```
.queryParams("userId", userId)
```

Loop Validation

Use a loop to validate every record instead of checking only one.

Example:

```
for (Integer id : userIds) {  
    assertEquals(userId, id);  
}
```

Expected Outcome

The automation should:

- Retrieve all users successfully.
- Select a userId dynamically.
- Fetch posts for the selected user.
- Verify the response is not empty.
- Confirm that every returned post belongs to the same userId.
- Pass all validations without errors.

Tools & Technologies

- Java
- Rest Assured
- JUnit 5
- JsonPath

Summary

This assessment demonstrates:

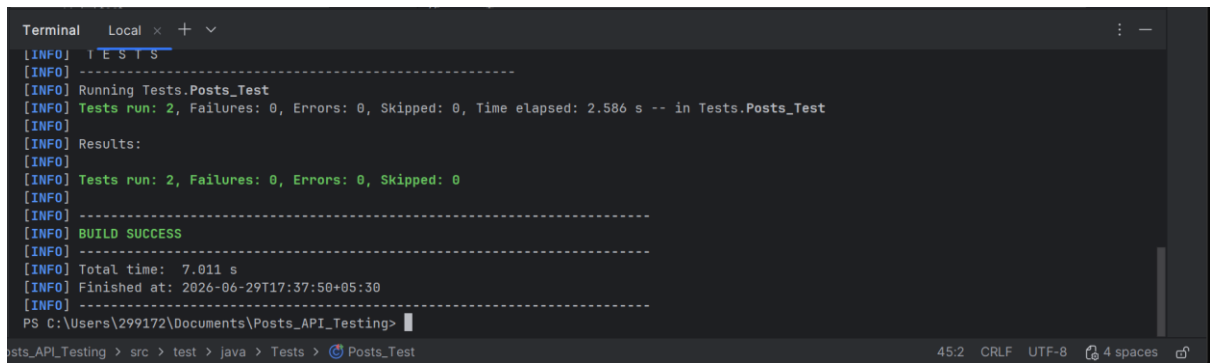
- REST API testing
- JSON parsing
- Dynamic data extraction

- Query parameter handling
- Loop-based validation
- Assertion of API responses
- Data consistency validation across multiple APIs

To Execute in the terminal:

Command: `mvn clean test`

Screenshot:



```
Terminal Local x + v
[INFO] T E S T S
[INFO] -----
[INFO] Running Tests.Posts_Test
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.586 s -- in Tests.Posts_Test
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.011 s
[INFO] Finished at: 2026-06-29T17:37:50+05:30
[INFO] -----
PS C:\Users\299172\Documents\Posts_API_Testing>
```

Posts_API_Testing > src > test > java > Tests > Posts_Test 45:2 CRLF UTF-8 4 spaces